

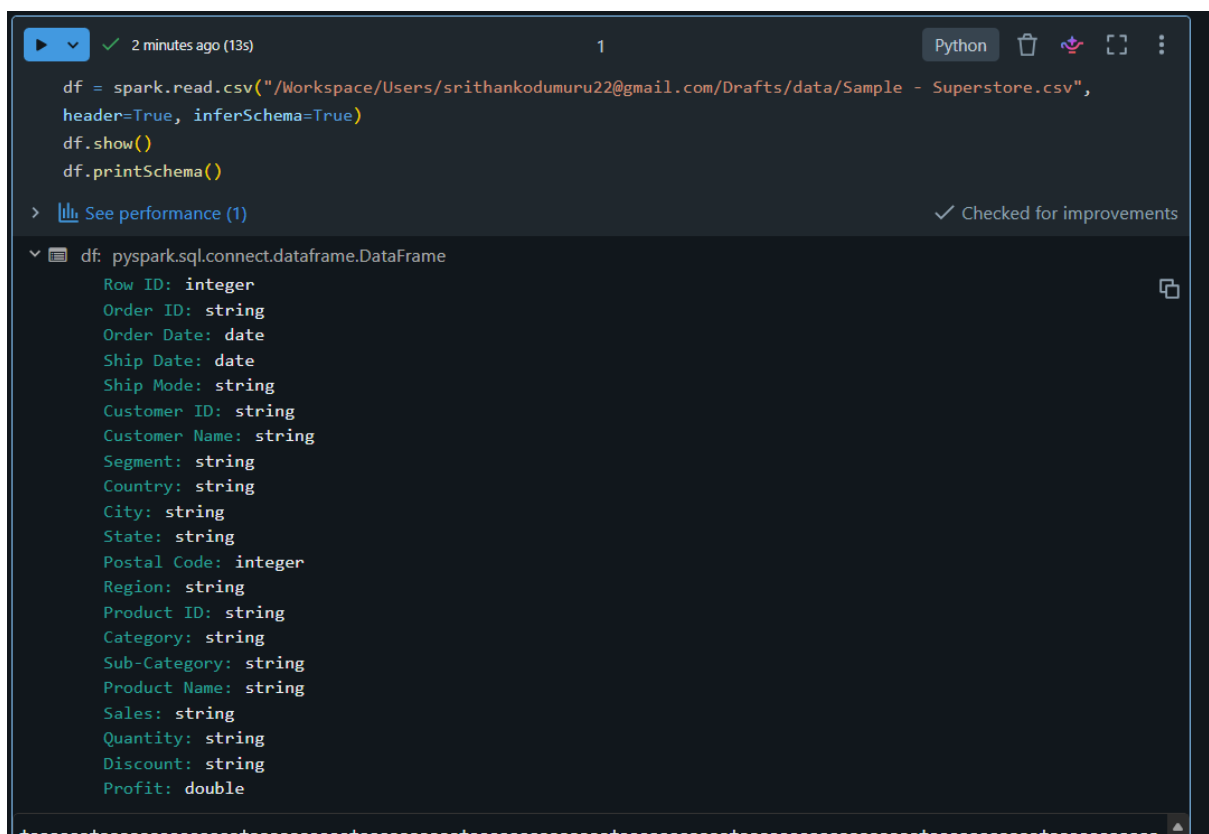
Q1. Explain the roles of the Driver, Cluster Manager, and Executor in a Spark application.

Answer: The Driver is the main part of a Spark application. It creates the Spark session, manages the program, and sends tasks to the executors. The Cluster Manager manages the cluster resources and assigns them to the Spark application. The Executors are the worker processes that run the tasks given by the Driver and process the data. They also store intermediate data during execution.

Q2. How does Spark's Lazy Evaluation strategy improve performance when chain-processing large datasets?

Answer: Spark does not execute the code immediately. It first stores all the transformations and waits until an action like `show()` or `collect()` is called. This helps Spark optimize the execution and avoid unnecessary work, making data processing faster and more efficient.

Q3: Write a Spark command to read a CSV file located at "data/source.csv", ensuring the first row is treated as a header and inferSchema is enabled.

A screenshot of a Jupyter Notebook interface. The top bar shows a play button, a checkmark, '2 minutes ago (13s)', a tab labeled '1', and a 'Python' language selector. The code cell contains:

```
df = spark.read.csv("/Workspace/Users/srithankodumuru22@gmail.com/Drafts/data/Sample - Superstore.csv",
header=True, inferSchema=True)
df.show()
df.printSchema()
```

Below the code, there is a link 'See performance (1)' and a status 'Checked for improvements'. The output area shows the schema for 'df: pyspark.sql.connect.dataframe.DataFrame':

```
Row ID: integer
Order ID: string
Order Date: date
Ship Date: date
Ship Mode: string
Customer ID: string
Customer Name: string
Segment: string
Country: string
City: string
State: string
Postal Code: integer
Region: string
Product ID: string
Category: string
Sub-Category: string
Product Name: string
Sales: string
Quantity: string
Discount: string
Profit: double
```

Q4. What is the difference between CSV and Parquet in terms of storage (row-based vs. columnar) and why does it matter for performance?

Answer : CSV stores data row by row, while Parquet stores data column by column. Parquet is faster because Spark reads only the required columns instead of the whole file. It also uses compression, which reduces storage space and improves performance.

Q5: Given a DataFrame df, write a query to select the columns product_id and price where the category is 'Electronics'.

```

▶ Just now (2s) 2 Python
df.select("Product ID","Sales")\
  .filter(df["Category"]=="Technology")\
  .show()
> See performance (1) ✓ Checked for improvements
+-----+-----+
| Product ID | Sales |
+-----+-----+
| TEC-PH-10002275 | 907.152 |
| TEC-PH-10002033 | 911.424 |
| TEC-PH-10001949 | 213.48 |
| TEC-AC-10003027 | 90.57 |
| TEC-PH-10004977 | 1097.544 |
| TEC-PH-10000486 | 371.168 |
| TEC-PH-10004093 | 147.168 |
| TEC-AC-10000171 | 45.98 |
| TEC-AC-10002167 | 45 |
| TEC-PH-10003988 | 21.8 |
| TEC-PH-10002447 | 1029.95 |
| TEC-AC-10002167 | 30 |
| TEC-AC-10004633 | 13.98 |
| TEC-PH-10002726 | 167.968 |
| TEC-AC-10001998 | 19.99 |
| TEC-PH-10004093 | 73.584 |
| TEC-AC-10001767 | 95.976 |
| TEC-AC-10001552 | 238.896 |

```

Note: Superstore dataset does not have an Electronics category. It uses categories like Technology, Furniture, and Office Supplies, so Technology is the correct value to filter.

Q6. Write the code to revise a DataFrame by renaming the column `old_name` to `new_name` and casting the price column from a String to a Double.

```

▶ 1 minute ago (2s) 3 Python
from pyspark.sql.functions import col
df=df.withColumnRenamed("Sales","Total Sales")\
  .withColumn("Total Sales",col("Total Sales").try_cast("double"))
df.show(5,truncate=False)
> See performance (1) ✓ Checked for improvements
> df: pyspark.sql.connect.dataframe.DataFrame = [Row ID: integer, Order ID: string ... 19 more fields]
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|Row ID|Order ID|Order Date|Ship Date|Ship Mode|Customer ID|Customer Name|Segment|Country|City|
|Total Sales|Quantity|Discount|Profit|Product ID|Category|Sub-Category|Product Name|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|1|CA-2016-152156|2016-11-08|2016-11-11|Second Class|CG-12520|Claire Gute|Consumer|United States|He
nderson|Kentucky|42420|South|FUR-B0-10001798|Furniture|Bookcases|Bush Somerset Collection
Bookcase|261.96|2|0|41.9136|
|2|CA-2016-152156|2016-11-08|2016-11-11|Second Class|CG-12520|Claire Gute|Consumer|United States|He
nderson|Kentucky|42420|South|FUR-CH-10000454|Furniture|Chairs|Hon Deluxe Fabric Upholst
ered Stacking Chairs, Rounded Back|731.94|3|0|219.582|
|3|CA-2016-138688|2016-06-12|2016-06-16|Second Class|DV-13045|Darrin Van Huff|Corporate|United States|Lo
s Angeles|California|90036|West|OFF-LA-10000240|Office Supplies|Labels|Self-Adhesive Address Lab
els for Typewriters by Universal|14.62|2|0|6.8714|
|4|US-2015-108966|2015-10-11|2015-10-18|Standard Class|SO-20335|Sean O'Donnell|Consumer|United States|Fo
rt Lauderdale|Florida|33311|South|FUR-TA-10000577|Furniture|Tables|Bretford CR4500 Series SI
m Rectangular Table|957.5775|5|0.45|-383.031|

```

Q7. How does Spark use the Lineage Graph (DAG) to provide fault tolerance if a worker node fails?

Answer : Spark keeps a record of all the transformations in a Lineage Graph (DAG). If a worker node fails and some data is lost, Spark uses this graph to recompute only the missing data instead of processing the entire dataset again. This provides fault tolerance and saves time.

Q8. Write a query to filter a DataFrame df_orders for rows where the status is 'Completed' AND the amount is greater than 1000.

```
from pyspark.sql.functions import col
df.filter((col("Region")=="South")&(col("Total Sales")>1000))\
.show()
```

> [See performance \(1\)](#) ✓ Checked for improvements

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Cou
ntry	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product
Name	Total Sales	Quantity	Discount	Profit				
510	CA-2015-145352	2015-03-16	2015-03-22	Standard Class	CM-12385	Christopher Martinez	Consumer	United St
ates	Atlanta	Georgia	30318	South	OFF-BI-10003527	Office Supplies	Binders	Fellowes PB500
El...	6354.95	5	0	3177.475				
684	US-2017-168116	2017-11-04	2017-11-04	Same Day	GT-14635	Grant Thornton	Corporate	United St
ates	Burlington	North Carolina	27217	South	TEC-MA-10004125	Technology	Machines	Cubify CubeX 3D
P...	7999.98	4	0.5	-3839.9904				
793	CA-2016-105256	2016-05-20	2016-05-20	Same Day	JK-15730	Joe Kamberova	Consumer	United St
ates	Asheville	North Carolina	28806	South	TEC-PH-10001530	Technology	Phones	Cisco Unified I
P...	1363.96	5	0.2	85.2475				
991	CA-2015-153549	2015-03-29	2015-03-31	Second Class	SL-20155	Sara Luxemburg	Home Office	United St
ates	Jacksonville	Florida	32216	South	FUR-CH-10004086	Furniture	Chairs	Hon 4070 Series
P...	1166.92	5	0.2	131.2785				

Q9. Explain the concept of Predicate Pushdown in Parquet and how it affects the amount of data loaded into memory.

Answer: Predicate Pushdown is a feature in Parquet where Spark applies filter conditions while reading the file. It reads only the required data instead of the entire dataset. This reduces memory usage, decreases disk I/O, and improves performance.

Q10. Write a code snippet to add a new column final_price which is the base_price multiplied by 1.18 (18% tax).

```

Just now (2s) 5 Python
from pyspark.sql.functions import col

df=df.withColumn("Final Price",col("Total Sales")*1.18)
df.show(5)

> See performance (1) ✓ Checked for improvements
> df: pyspark.sql.connect.dataframe.DataFrame = [Row ID: integer, Order ID: string ... 20 more fields]
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|Row ID|  Order ID|Order Date| Ship Date|  Ship Mode|Customer ID| Customer Name| Segment|  Country|
City|  State|Postal Code|Region|  Product ID|  Category|Sub-Category|  Product Name|Total Sales|Q
quantity|Discount| Profit|  Final Price|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1|CA-2016-152156|2016-11-08|2016-11-11| Second Class| CG-12520| Claire Gute| Consumer|United States|
Henderson| Kentucky| 42420| South|FUR-BO-10001798| Furniture| Bookcases|Bush Somerset Col...| 26
1.96| 2| 0| 41.9136|309.11279999999994|
| 2|CA-2016-152156|2016-11-08|2016-11-11| Second Class| CG-12520| Claire Gute| Consumer|United States|
Henderson| Kentucky| 42420| South|FUR-CH-10000454| Furniture| Chairs|Hon Deluxe Fabric...| 73
1.94| 3| 0| 219.582| 863.6892|
| 3|CA-2016-138688|2016-06-12|2016-06-16| Second Class| DV-13045|Darrin Van Huff|Corporate|United States|
Los Angeles|California| 90036| West|OFF-LA-10000240|Office Supplies| Labels|Self-Adhesive Add...|
14.62| 2| 0| 6.8714| 17.2516|
| 4|US-2015-108966|2015-10-11|2015-10-18|Standard Class| SO-20335| Sean O'Donnell| Consumer|United States|Fo
rt Lauderdale| Florida| 33311| South|FUR-TA-10000577| Furniture| Tables|Bretford CR4500 S...| 9
57.5775| 5| 0.45|-383.031| 1129.94145|

```

Q11. What is the difference between Transformations and Actions? Provide two examples of each.

Answer: Transformations are operations that change the data or create a new dataframe , but Spark does not execute them immediately. It waits until an action is called. Actions are operations that execute all the transformations and return the result.

Examples of Transformations: filter(), select()

Examples of Actions: show(), collect()

Q12. Write the Spark command to load a Parquet file from "path/to/input", filter out any rows where user_id is null, and save the result as a CSV at "path/to/output".

I completed all the Spark concepts and implemented the required code for the assignment. However, I was unable to execute the final write operation (saving the DataFrame as Parquet/CSV) because the Databricks workspace provided has storage restrictions.

While reading the dataset worked successfully, the write operation failed with a workspace/DBFS permission error (DBFS_DISABLED / unable to create directories). Due to this environment limitation, I could not generate the output files.

I have asked my mentor for the correct writable storage path or the recommended environment to complete this step. Once it is provided, I will execute the code and update the output.

Q13. In Spark Architecture, what is the difference between Client Mode and Cluster Mode?

Answer : In Client Mode, the Driver program runs on the local machine where the Spark application is submitted. In Cluster Mode, the Driver runs inside the cluster along with the executors. Client Mode is mainly used for development and testing, while Cluster Mode is preferred for running production applications.

Q14. Write a query to filter a dataset for rows where the region is 'North' OR the priority is 'High'.

```
from pyspark.sql.functions import col

df.filter((col("Region") == "North") | (col("Segment") == "Consumer")) \
.show()
```

> [See performance \(1\)](#) ✓ Checked for improvements

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country
City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Total Sales
Quantity	Discount	Profit	Final Price					
1	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States
Henderson	Kentucky	42420	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Col...	261.96
2	0	41.9136	309.11279999999994					
2	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States
Henderson	Kentucky	42420	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric...	731.94
3	0	219.582	863.6892					
4	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States
Fort Lauderdale	Florida	33311	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500	957.5775
S...	5	0.45	-383.031	1129.94145				
5	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States
Fort Lauderdale	Florida	33311	South	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold 'N Ro	22.368
1...	2	0.2	2.5164	26.394239999999996				

Q15. When exploring a dataset, why is it safer to use .show(5) instead of .collect() on a multi-terabyte dataset?

Answer : .show(5) displays only the first 5 rows of the dataset, so it uses very little memory. On the other hand, .collect() loads the entire dataset into the Driver's memory. If the dataset is very large, it can cause memory issues or even crash the application. That's why .show(5) is safer for exploring large datasets.

Workflow

Read CSV Dataset



Check Schema



Select Required Columns



Filter Data



Modify DataFrame



Handle Null Values



Save Output